SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

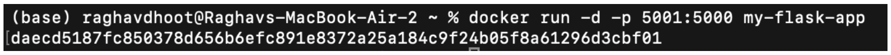
(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of india)

॥वसुधैव कुटुम्बकम्॥ Re-Accredited by NAAC with 'A' Grade

Department of Computer Science and Engineering
DevOps Lab - L3

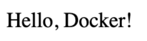
Lab Assignment No: 9	
Name of Student	Raghav Dhoot
PRN	23070122170
Batch	2023 - 2027
Class	CSE - B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Container Management & Networking

Implementation & Theory



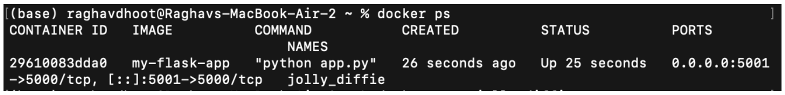
```
(base) raghavdhoot@Raghavs-MacBook-Air-2 ~ % docker run -d -p 5001:5000 my-flask-app
[daecd5187fc850378d656b6efc891e8372a25a184c9f24b05f8a61296d3cbf01
```

Run container detached — Started a new container from the my-flask-app image in detached mode (-d) with port mapping 5001:5000, returning the container ID.



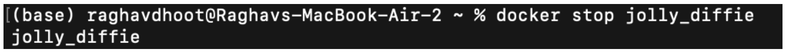
Hello, Docker!

Access via browser — Verified the app was reachable at http://localhost:5001, displaying "Hello, Docker!"



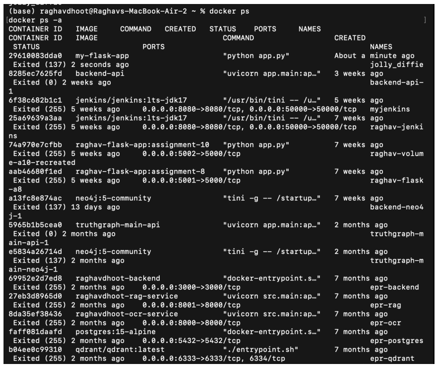
```
[(base) raghavdhoot@Raghavs-MacBook-Air-2 ~ % docker ps                                         ]
CONTAINER ID   IMAGE          COMMAND           CREATED          STATUS          PORTS
                                        NAMES
29610083dda0   my-flask-app   "python app.py"   26 seconds ago   Up 25 seconds   0.0.0.0:5001
->5000/tcp, [::]:5001->5000/tcp   jolly_diffie
```

List running containers — docker ps showed the container's status as "Up," with the port mapping 0.0.0.0:5001->5000/tcp confirmed.



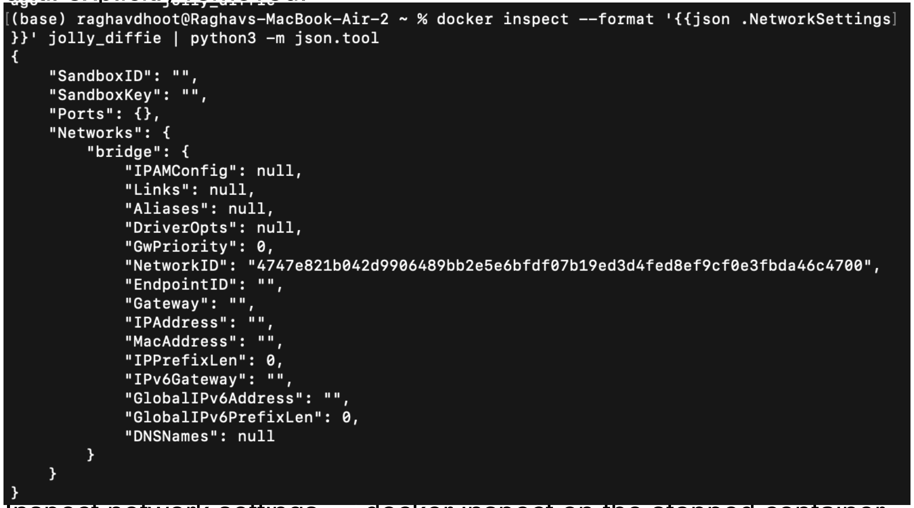
```
[(base) raghavdhoot@Raghavs-MacBook-Air-2 ~ % docker stop jolly_diffie
jolly_diffie
```

Stop the container — Used docker stop with the container ID/name; Docker returned the same identifier once stopped.



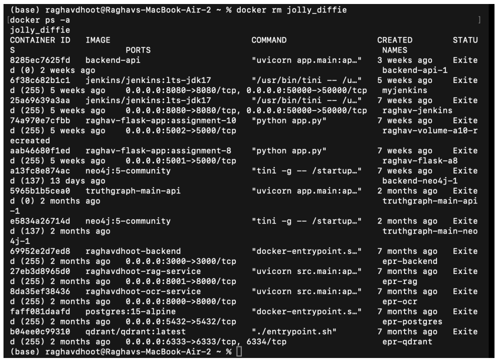
```
(base) raghavdhoot@Raghavs-MacBook-Air-2 ~ % docker ps
docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
CONTAINER ID   IMAGE                             COMMAND                  CREATED
 STATUS                    PORTS                                               NAMES
29610083dda0   my-flask-app                      "python app.py"          About a minute ago
 Exited (137) 2 seconds ago                                                    jolly_diffie
8285ec7625fd   backend-api                       "uvicorn app.main:ap…"   3 weeks ago
 Exited (0) 2 weeks ago                                                        backend-api-
1
6f38c682b1c1   jenkins/jenkins:lts-jdk17         "/usr/bin/tini -- /u…"   5 weeks ago
 Exited (255) 5 weeks ago     0.0.0.0:8080->8080/tcp, 0.0.0.0:50000->50000/tcp   myjenkins
25a69639a3aa   jenkins/jenkins:lts-jdk17         "/usr/bin/tini -- /u…"   7 weeks ago
 Exited (255) 5 weeks ago     0.0.0.0:8080->8080/tcp, 0.0.0.0:50000->50000/tcp   raghav-jenki
ns
74a970e7cfbb   raghav-flask-app:assignment-10   "python app.py"          7 weeks ago
 Exited (255) 5 weeks ago     0.0.0.0:5002->5000/tcp                           raghav-volum
e-a10-recreated
aab46680f1ed   raghav-flask-app:assignment-8    "python app.py"          7 weeks ago
 Exited (255) 5 weeks ago     0.0.0.0:5001->5000/tcp                           raghav-flask
-a8
a13fc8e874ac   neo4j:5-community                "tini -g -- /startup…"   7 weeks ago
 Exited (137) 13 days ago                                                      backend-neo4
j-1
5965b1b5cea0   truthgraph-main-api              "uvicorn app.main:ap…"   2 months ago
 Exited (0) 2 months ago                                                       truthgraph-m
ain-api-1
e5834a26714d   neo4j:5-community                "tini -g -- /startup…"   2 months ago
 Exited (137) 2 months ago                                                     truthgraph-m
ain-neo4j-1
69952e2d7ed8   raghavdhoot-backend              "docker-entrypoint.s…"   7 months ago
 Exited (255) 2 months ago    0.0.0.0:3000->3000/tcp                           epr-backend
27eb3d8965d0   raghavdhoot-rag-service          "uvicorn src.main:ap…"   7 months ago
 Exited (255) 2 months ago    0.0.0.0:8001->8000/tcp                           epr-rag
8da35ef38436   raghavdhoot-ocr-service          "uvicorn src.main:ap…"   7 months ago
 Exited (255) 2 months ago    0.0.0.0:8000->8000/tcp                           epr-ocr
faff081daafd   postgres:15-alpine               "docker-entrypoint.s…"   7 months ago
 Exited (255) 2 months ago    0.0.0.0:5432->5432/tcp                           epr-postgres
b04ee0c99310   qdrant/qdrant:latest             "./entrypoint.sh"        7 months ago
 Exited (255) 2 months ago    0.0.0.0:6333->6333/tcp, 6334/tcp                 epr-qdrant
```

Running vs. all containers — docker ps returned an empty list (no running containers), while docker ps -a showed the container still present with status "Exited," confirming stopped containers persist until explicitly removed.



```
(base) raghavdhoot@Raghavs-MacBook-Air-2 ~ % docker inspect --format '{{json .NetworkSettings
}}' jolly_diffie | python3 -m json.tool
{
    "SandboxID": "",
    "SandboxKey": "",
    "Ports": {},
    "Networks": {
        "bridge": {
            "IPAMConfig": null,
            "Links": null,
            "Aliases": null,
            "DriverOpts": null,
            "GwPriority": 0,
            "NetworkID": "4747e821b042d9906489bb2e5e6bfdf07b19ed3d4fed8ef9cf0e3fbda46c4700",
            "EndpointID": "",
            "Gateway": "",
            "IPAddress": "",
            "MacAddress": "",
            "IPPrefixLen": 0,
            "IPv6Gateway": "",
            "GlobalIPv6Address": "",
            "GlobalIPv6PrefixLen": 0,
            "DNSNames": null
        }
    }
}
```

Inspect network settings — docker inspect on the stopped container revealed the NetworkSettings block, including its internal IP address, gateway, and configured port bindings.



```
(base) raghavdhoot@Raghavs-MacBook-Air-2 ~ % docker rm jolly_diffie
docker ps -a
jolly_diffie
CONTAINER ID   IMAGE                             COMMAND                  CREATED         STATU
S                       PORTS                                               NAMES
8285ec7625fd   backend-api                       "uvicorn app.main:ap…"   3 weeks ago     Exite
d (0) 2 weeks ago                                                           backend-api-1
6f38c682b1c1   jenkins/jenkins:lts-jdk17         "/usr/bin/tini -- /u…"   5 weeks ago     Exite
d (255) 5 weeks ago    0.0.0.0:8080->8080/tcp, 0.0.0.0:50000->50000/tcp   myjenkins
25a69639a3aa   jenkins/jenkins:lts-jdk17         "/usr/bin/tini -- /u…"   7 weeks ago     Exite
d (255) 5 weeks ago    0.0.0.0:8080->8080/tcp, 0.0.0.0:50000->50000/tcp   raghav-jenkins
74a970e7cfbb   raghav-flask-app:assignment-10   "python app.py"          7 weeks ago     Exite
d (255) 5 weeks ago    0.0.0.0:5002->5000/tcp                             raghav-volume-a10-r
ecreated
aab46680f1ed   raghav-flask-app:assignment-8    "python app.py"          7 weeks ago     Exite
d (255) 5 weeks ago    0.0.0.0:5001->5000/tcp                             raghav-flask-a8
a13fc8e874ac   neo4j:5-community                "tini -g -- /startup…"   7 weeks ago     Exite
d (137) 13 days ago                                                         backend-neo4j-1
5965b1b5cea0   truthgraph-main-api              "uvicorn app.main:ap…"   2 months ago    Exite
d (0) 2 months ago                                                          truthgraph-main-api
-1
e5834a26714d   neo4j:5-community                "tini -g -- /startup…"   2 months ago    Exite
d (137) 2 months ago                                                        truthgraph-main-neo
4j-1
69952e2d7ed8   raghavdhoot-backend              "docker-entrypoint.s…"   7 months ago    Exite
d (255) 2 months ago   0.0.0.0:3000->3000/tcp                             epr-backend
27eb3d8965d0   raghavdhoot-rag-service          "uvicorn src.main:ap…"   7 months ago    Exite
d (255) 2 months ago   0.0.0.0:8001->8000/tcp                             epr-rag
8da35ef38436   raghavdhoot-ocr-service          "uvicorn src.main:ap…"   7 months ago    Exite
d (255) 2 months ago   0.0.0.0:8000->8000/tcp                             epr-ocr
faff081daafd   postgres:15-alpine               "docker-entrypoint.s…"   7 months ago    Exite
d (255) 2 months ago   0.0.0.0:5432->5432/tcp                             epr-postgres
b04ee0c99310   qdrant/qdrant:latest             "./entrypoint.sh"        7 months ago    Exite
d (255) 2 months ago   0.0.0.0:6333->6333/tcp, 6334/tcp                   epr-qdrant
(base) raghavdhoot@Raghavs-MacBook-Air-2 ~ %
```

Remove the container — docker rm deleted the stopped container; a follow-up docker ps -a confirmed it no longer appears in the container list.